Power Propagation and Frequency Coupling Characteristics of Wind Turbine Multi-Physical System With AC Grid

Lulan Yin, Weihao Hu, Rongwu Zhu, *Senior Member, IEEE*, Marco Liserre, *Fellow, IEEE*, and Zhe Chen, *Fellow, IEEE*

***Abstract*—Industrial field reports show that wind turbine (WT)-based energy systems are prone to result in power grids suffering from oscillating problems. One of the main differences between the WT system and the photovoltaic system is the property of their primary energy, the former is rotating and the latter is static. Hence, in this paper, the multi-physical domain modeling of a doubly-fed induction generator (DFIG)-based wind turbine system, composed of aerodynamics, mechanics, electromagnetism, power electronics and control system, is derived to study the power propagation and frequency coupling characteristics from wind energy to electricity grids, showing the impacts of the WT system on electricity grid performances. Theoretical analysis shows that the $3np$ Hz distortion in wind energy can be converted to $50 \pm 3np$ Hz oscillations in AC grids. The impacts of different control schemes (power control and speed control) of DFIG are studied in detail to show that the control schemes of the back-to-back converter play a significant role in influencing the power propagation characteristics. The results of PLECS-based simulation and OPAL-RT-based test clearly validate the correctness of the theoretical analyzes.**

***Index Terms*—Wind energy, power propagation, frequency coupling, drive train, back-to-back converter, doubly-fed induction generator (DFIG), $50 \pm 3np$ Hz oscillations.**

I. Introduction

With continuously increasing the installed capacity of renewable energy, e.g., wind power and solar power, the modern electricity grid experiences a high penetration of power electronics-interfaced renewable energy resources (RESs) [1]. By the year of 2023, the installed capacity of RESs in China had reached 1.45 billion kW, sharing more than 50% of the total installed capacity of the country. However, the high penetration of RESs results in the modern power system suffering from the issues of oscillations and voltage violations [2].

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Lulan Yin, Weihao Hu, and Rongwu Zhu are with the Department of Electrical Engineering, Harbin Institute of Technology, Shenzhen 518055, China (e-mail: 22s053054@stu.hit.edu.cn; 23S053052@stu.hit.edu.cn; rzh@hit.edu.cn).

Marco Liserre is with the Chair of Power Electronics, Kiel University, 24143 Kiel, Germany (e-mail: ml@tf.uni-kiel.de).

Zhe Chen is with the Department of Energy Technology, Aalborg University, 9220 Aalborg, Denmark (e-mail: zch@et.aau.dk).

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The issue of sub-synchronous oscillations caused by RESs was primarily occurred in the power grids with a high penetration of wind energy, e.g. the oscillating events reported in wind farms in USA, Europe, Austria, and China. Although the grid-interfaced inverters of the wind turbine (WT) system and the photovoltaic system are similar, one of the main differences between them is the property of their primary energy; the former is rotating and the latter is static. Hence, it is essential to study the characteristics of power propagation from wind energy to electricity grids, revealing the impacts of the WT system on the performance of power grids. Apart from the interfaced coupling between the grid-tied power converters and the ac grid, the impacts of the drive train system (blade, hub, gearbox and generator) of WTs should also be studied.

The WT may considerably affect the supplied power quality by the power propagation characteristics in the drive train system and the back-to-back converter. It comes from the fluctuating characteristics of their output power, which contains the periodic components because of the aerodynamic phenomena, e.g. the wind shear and tower shadow effects. The term wind shear is used to describe the variation in wind speed with height, while the tower shadow describes the reduction of wind velocity on the leeward side of the tower [3]. In three-bladed turbines, the primary and most substantial periodic power pulsations caused by wind shear and tower shadow occur at a frequency denoted as 3p [4], which is three times the rotor frequency. Thus, even under conditions of constant wind speed at a specific height, the turbine blade experiences a varying wind speed when rotating. Consequently, wind power pulsations and aerodynamic torque pulsations are observed. The pulsations are then propagated through the drive train and then transformed into the electromagnetic torque of generator through the WT drive train system, as shown in Fig. 1, and then result in disturbances in the electrical system [5], [6]. Besides the physical system, the control schemes of the generator have significant impacts on the propagation characteristics of the fluctuating power. Hence, the propagation characteristics of the drive train and back-to-back converter are studied in detail in this paper.

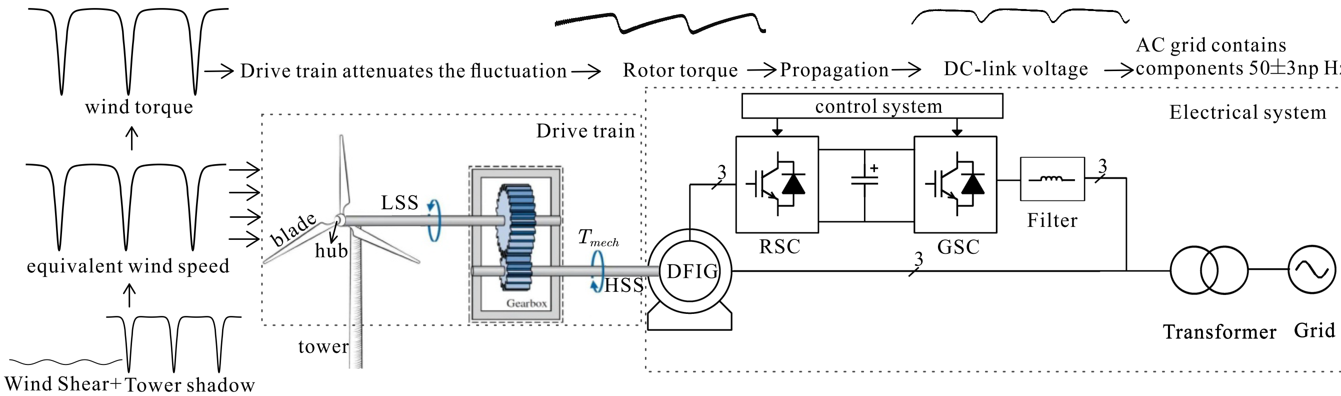


Fig. 1. Overall system of DFIG-based wind turbine.

Previous studies indicate that power fluctuations of WTs caused by wind shear and tower shadow effects can cause flicker [7], [8]. However, while most studies address the time-domain fluctuations of wind power and electromagnetic power separately through simulations [9], [10], and some approximate the fluctuations of electromagnetic power as being equivalent to those of wind power [11]. There remains a lack of quantitative analysis on the propagation of fluctuating power and its impact on the performance of AC grids. Besides, dynamic models of WTs had been studied, but the aerodynamic torque was simply represented by a steady-state torque curve under the spatial mean wind speed [10], [12] or sinusoidal oscillations [13]. These approximations can not accurately reflect the authentic fluctuations in the aerodynamic torque of the WT, and the dynamic behavior of the drive train was often ignored [14], [15] or simply presented by an one-mass model [16], which is unable to reflect the torsional characteristics and transient responses of the drive train [17]. These models are insufficient to study the periodic power oscillations of WT systems [18]. This paper provides a detailed modeling of the multi-physical power propagation route, and further quantitatively analyzes the power propagation and frequency coupling characteristics form wind energy to electricity grids.

The control scheme affects the power propagation and frequency coupling characteristics of DFIG-based WTs. Many studies focus on control strategies at various wind speeds [19], [20], and some also analyze the stability issue using established models [21], [22]. However, few works focus on the influence of the control scheme on the propagation characteristics of wind power pulsations, and oscillation frequencies were studied through simulations [7], [22]. Hence, this paper quantitatively investigates the power propagation characteristics affected by the control system, along with a theoretical analysis of the frequency coupling characteristics and the oscillation frequencies on the grid side.

This paper focuses on the power propagation characteristics from wind energy to electricity grids based on the multi-physical model of wind energy system. The main contributions of this paper are summarized as follows.

1) A detailed multi-physical domain model of the DFIG-based WT system, composed of aerodynamical system, mechanical system, generator, and power electronics converters as well as their control system, is derived.
2) The power propagation characteristics from fluctuating wind power to electromagnetic power are quantitatively analyzed, considering the impacts of the mechanical drive train system and control schemes of the WT system, and the frequency coupling characteristics between power fluctuations and grid current oscillations are investigated, revealing that $3np$ Hz distortion in wind energy can be converted to $50 \pm 3np$ Hz oscillation in the AC grid.
3) The effects of different control schemes (stator active power control and speed control) on the power propagation characteristics are studied in detail. Under the active power control scheme, the fluctuating wind power across all frequencies can be buffered and stored in the mechanical system, without directly propagating to the electrical system, resulting in less impact on AC grid performances. In contrast, under the speed control scheme, low-frequency fluctuating wind power is propagated into the electrical grid with minimal attenuation or even slight amplification, leading to frequency coupling issues.

The rest of this paper is organized as follows. In Section II, the mechanical system of the WT drive train is modeled as a six-mass model and then converted into an equivalent two-mass model, which is effective to study the dynamic behavior of wind shear and tower shadow, and the electrical system of generator and back-to-back converter are modeled as well. In Section III, the impacts of multi-physical system on the power propagation characteristics are analyzed. In Section IV, based on derived model of drive train, the impacts of control schemes (power control and speed control) on the electricity grid performances are analyzed in detail. In Section V, the simulation results carried out in PLECS clearly validate the correctness of the theoretical analyses. In Section VI, the OPAL-RT based test results are conducted. This paper is concluded in Section VII.

II. Multi-Physical Models of WT System

A. Overall System of DFIG-Based WT

The overall system of DFIG-based wind energy system consisting of aerodynamical system, mechanical drive train system and electrical system is shown in Fig. 1. The equivalent wind speed, wind power and aerodynamic torque of WT fluctuate at 3p frequency due to the effects of wind shear and tower shadow. The mechanical drive train system is composed of 3 blades, hub, gearbox, generator, and shaft. The blades are in charge of absorbing and transforming the energy from the wind into mechanical torque and power, which have the same fluctuation frequency as the wind power. The electrical system is composed of DFIG, back-to-back converter, filter, transformer, and grid, and the control scheme of the back-to-back converter impacts the propagation of fluctuating electromagnetic power to electricity grids.

B. Dynamic Model of WT Drive Train

To obtain a more accurate aerodynamic torque input for the drive train, an equivalent wind speed of WT, accounting for the effects of wind shear and tower shadow, is expressed as,

$$V_{\mathrm{eq}} = V_{\mathrm{H}} + V_{\mathrm{eq-ws}} + V_{\mathrm{eq-ts}}, \tag{1}$$

where V_{H} is the hub height wind speed, $V_{\mathrm{eq-ws}}$ and $V_{\mathrm{eq-ts}}$ are the equivalent wind speed due to wind shear and tower shadow, respectively.

The formulas of $V_{\mathrm{eq-ws}}$ and $V_{\mathrm{eq-ts}}$ in [3] are expressed as,

$$V_{\mathrm{eq-ws}} = V_{\mathrm{H}}\alpha\left(\alpha-1\right)\left(\frac{R}{H}\right)^2\left[\frac{1}{8}+\frac{\left(\alpha-2\right)}{60}\frac{R}{H}\cos 3\theta\right], \tag{2}$$

$$V_{\mathrm{eq-ts}} = \frac{mV_{\mathrm{H}}}{3R^2}\sum_{\mathrm{b}=1}^{3}\left[\frac{a^2}{\sin^2\theta_{\mathrm{b}}}\ln\left(\frac{R^2\sin^2\theta_{\mathrm{b}}}{x^2}+1\right) - \frac{2a^2R^2}{R^2\sin^2\theta_{\mathrm{b}}+x^2}\right], \tag{3}$$

where α is the empirical wind shear exponent, θ_{b} and θ are the azimuth angle of blades, *a* is the tower radius, *x* is the blade origin from tower midline, *H* is the hub height, *R* is the turbine radius and *m* is the ratio of spatial average wind speed to hub height wind speed, which can be computed by,

$$m = 1 + \frac{\alpha\left(\alpha-1\right)R^2}{8H^2}. \tag{4}$$

The wind power P_{w} and aerodynamical torque T_{w} extracted by blades can be computed by,

$$\begin{cases} P_{\mathrm{w}} = \frac{1}{2}\rho A C_{\mathrm{p}} V_{\mathrm{eq}}^3, \\ T_{\mathrm{w}} = \frac{P_{\mathrm{w}}}{\Omega_{\mathrm{wt}}}, \end{cases} \tag{5}$$

where ρ is the air density, *A* is the area swept by WT blades, C_{p} is the performance coefficient of WT, and Ω_{wt} is the mechanical rotational speed of WT.

By substituting (1)-(4) into (5), it can be obtained that the wind power contains the constant component $\bar{P}_{\mathrm{w}}$ and the fluctuation component $\tilde{P}_{\mathrm{w}}$, as shown in (6). Based on (5), when Ω_{wt} is constant, the fluctuation frequencies of $\tilde{P}_{\mathrm{w}}$ and $\tilde{T}_{\mathrm{w}}$ are the same as that of V_{eq}, and equal to the blade passing frequency, also known as 3 times the rotor frequency (3p).

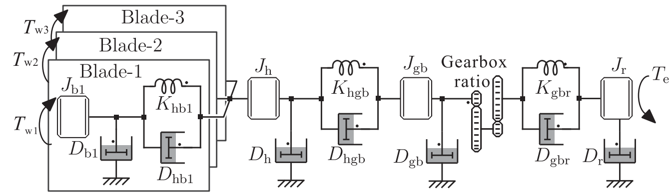


Fig. 2. Six-mass model of WT mechanical system.

$$\begin{cases} P_{\mathrm{w}} = \bar{P}_{\mathrm{w}} + \tilde{P}_{\mathrm{w}}(3n\mathrm{p}), \\ T_{\mathrm{w}} = \bar{T}_{\mathrm{w}} + \tilde{T}_{\mathrm{w}}(3n\mathrm{p}), \end{cases} \tag{6}$$

where superscripts "¯" and "~" represent the constant and 3p fluctuating components, respectively.

The detailed model of drive train system through six rotating masses by shafts is illustrated in Fig. 2, where the model is comprised of six lumped inertia corresponding to the three blades, hub, gearbox, and generator rotor, along with coupling springs that represent the shaft connections between them [23].

The linearized dynamic equations of mechanical system in Fig. 2 are given by,

$$\begin{cases} \Delta T_j + \Delta T_{ij} - \Delta T_{jk} = J_j\frac{d\Delta\omega_j}{dt} + D_j\Delta\omega_j, \\ \Delta T_{ij} = K_{ij}\Delta\theta_{ij} + D_{ij}\left(\Delta\omega_i - \Delta\omega_j\right), \\ \frac{d\Delta\theta_{ij}}{dt} = \Delta\omega_i - \Delta\omega_j, \end{cases} \tag{7}$$

where Δ represents small deviation near the the steady-state operating point; subscripts "i", "j", and "k" represent the variables of mass-i, -j and -k, respectively; subscripts "ij" and "jk" represent the coupling variables between mass-i and -j, and between mass-j and -k, respectively; T, D, and J are the torque, damping coefficient and moment of inertia; ω is the angular speed; K is the stiffness coefficient of shaft.

It is worth noting that the dynamic equations for reduced order drive train system models can also be represented by (7). The small signal model of the generator electromagnetic torque and angular displacement can be given by [24],

$$\Delta T_{\mathrm{e}} = K_{\mathrm{e}}\Delta\theta_{\mathrm{r}}. \tag{8}$$

Based on (1)-(8), the drive train system can be modelled in the matrix form.

$$\boldsymbol{A}\boldsymbol{\Delta}\ddot{\boldsymbol{\theta}} + \boldsymbol{B}\boldsymbol{\Delta}\dot{\boldsymbol{\theta}} + \boldsymbol{C}\boldsymbol{\Delta}\boldsymbol{\theta} = \boldsymbol{\Delta T}, \tag{9}$$

where $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$ are the matrices of inertia, damping, and stiffness of the six-mass drive train system; $\boldsymbol{\Delta}\ddot{\boldsymbol{\theta}}$, $\boldsymbol{\Delta}\dot{\boldsymbol{\theta}}$, and $\boldsymbol{\Delta\theta}$ are the vectors of angular acceleration, velocity, and displacement of the system; and $\boldsymbol{\Delta T}$ is the torque vector.

To analyze the impacts of mechanical drive train system on the power propagation characteristics, it is necessary to convert the six-mass drive train model into a lower-order model. In the transformation procedure as follows, a simplified two-mass model is adopted, which is sufficient enough for the transient behavior analysis of drive train, but with much less complexity [25], [26].

Based on Table I, due to the lower stiffness coefficient of the low-speed shaft compared to the high-speed shaft, the blades and

TABLE I
SYSTEM PARAMETERS OF WT DRIVE TRAIN AND GENERATOR [29]

Parameters	Symbols	Values
Hub height wind speed (m/s)	V_{H}	11.2
Tower radius (m)	a	2.1
Empirical wind shear exponent	α	0.3
Blade origin from tower midline (m)	x	4
Hub height (m)	H	72.2
Turbine radius (m)	R	36.1
Rotation period of the blade (s)	t_{p}	2.5
Gearbox ratio	n_{g}	62
Rated output power (MW)	P_{n}	2
The number of pole pairs	n_{p}	2
Base angular electrical frequency (rad/s)	ω_0	100π
Stator resistance (Ω)	R_{s}	0.022
Rator resistance (Ω)	R_{r}	0.0018
Stator inductance (mH)	L_{s}	3.02
Rotor inductance (mH)	L_{r}	2.95
Mutual inductance (mH)	L_{m}	2.9
Slip	s_{l}	0
Rotor inertia (kgm^2)	J_{r}	75
Gearbox inertia (kgm^2)	J_{gb}	1.66×10^5
Hub inertia (kgm^2)	J_{h}	2.35×10^4
Blade inertia (kgm^2)	J_{b}	4.39×10^5
Rotor friction (Nms/rad)	D_{r}	0.81
Gearbox friction (Nms/rad)	D_{gb}	6.96×10^3
Hub friction (Nms/rad)	D_{h}	3.16×10^3
Blade friction (Nms/rad)	D_{b}	4.22×10^2
Gearbox to rotor stiffness (Nm/rad)	K_{gbr}	4.67×10^7
Hub to gearbox stiffness (Nm/rad)	K_{hgb}	5.44×10^9
Blade to hub stiffness (Nm/rad)	K_{hb}	4.17×10^{10}
Gearbox to rotor damping (Nms/rad)	D_{gbr}	810
Hub to gearbox damping (Nms/rad)	D_{hgb}	1.11×10^6
Blade to hub damping (Nms/rad)	D_{hb}	1.27×10^6

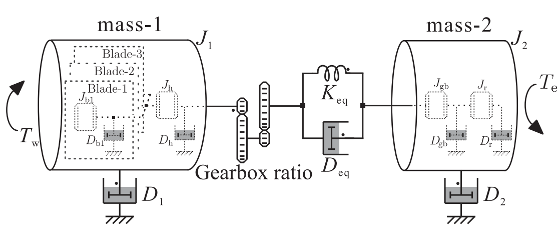


Fig. 3. Equivalent two-mass model of WT drive train.

hub at low-speed side, and the gearbox and generator at high-speed side, should be added into two different masses separately when the flexibility as well as damping of the shafts between them can be neglected [26], as shown in Fig. 3, where J_1 and J_2, as well as D_1 and D_2 represent the moment of inertia and self-damping of the equivalent mass-1 and mass-2, respectively. They are obtained by adding together J_{b} and J_{h} for J_1, J_{gb} and J_{r} for J_2, D_{b} and D_{h} for D_1, and D_{gb} and D_{r} for D_2.

The two masses are connected with the equivalent shaft stiffness K_{eq} and mutual-damping D_{eq}, which can be calculated from [27],

$$\begin{cases} \frac{1}{K_{\mathrm{eq}}} = \frac{n_{\mathrm{g}}^2}{K_{\mathrm{hgb}}} + \frac{1}{K_{\mathrm{gbr}}}, \\ \frac{1}{D_{\mathrm{eq}}} = \frac{n_{\mathrm{g}}^2}{D_{\mathrm{hgb}}} + \frac{1}{D_{\mathrm{gbr}}}, \end{cases} \tag{10}$$

where n_{g} is the gearbox ratio, K_{hgb} and D_{hgb} are the hub to gearbox stiffness and damping, K_{gbr} and D_{gbr} are the gearbox to generator rotor stiffness and damping, respectively.

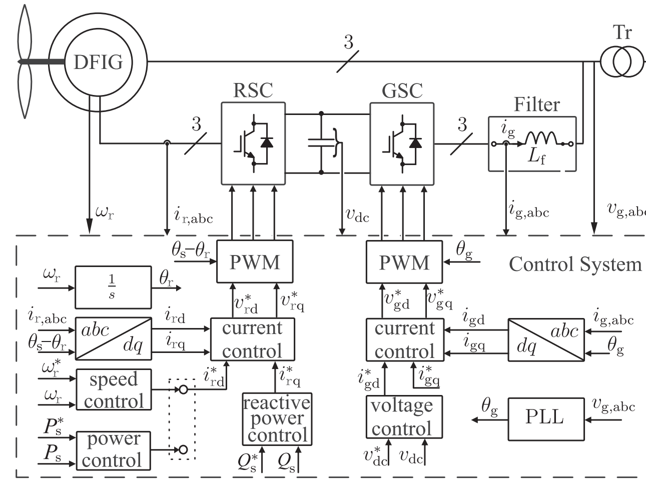


Fig. 4. Control block diagram of the RSC and GSC.

C. Electric System Model of DFIG-Based WT

The electrical system of DFIG-based WT is composed of DFIG, back-to-back converter and AC filter, and the details of overall control block diagram of the rotor-side converter (RSC) and grid-side converter (GSC) are shown in Fig. 4. The vector control method [28] is used to control the RSC and GSC.

Apart from the stator reactive power control, another control objective of the RSC converter is to control either the stator power or the rotor speed of DFIG. The control system of RSC is based on the cascaded loops, where the inner loop is the rotor current control, while the outer loop is either the rotor speed and stator reactive power control or the stator active and reactive power control. All the variables are aligned with the stator voltage orientation ($u_{\mathrm{sd}} = U_s,\ u_{\mathrm{sq}} = 0$), and the rotor current loops are decoupled into the d- and q-axis components, which are also called as active and reactive current components, respectively.

When the control objectives of RSC are the rotor speed and stator reactive power, the references of d- and q-axis components of rotor current are obtained by,

$$\begin{cases} i_{\mathrm{r-d}}^* = \left(k_{\mathrm{p}\omega} + \frac{k_{\mathrm{i}\omega}}{s}\right)\left(\omega_{\mathrm{r}}^* - \omega_{\mathrm{r}}\right), \\ i_{\mathrm{r-q}}^* = -\left(k_{\mathrm{pQ}} + \frac{k_{\mathrm{iQ}}}{s}\right)\left(Q_{\mathrm{s}}^* - Q_{\mathrm{s}}\right). \end{cases} \tag{11}$$

where superscript "*" represents the reference, k_{p} and k_{i} represent the proportional gain and integral gain, respectively, subscripts "ω" and "Q" represent the variables of speed and stator reactive power control loops, respectively, ω_{r} is the rotor angular frequency, Q_{s} is the stator reactive power.

When the control objectives are the stator active and reactive power, the references of d- and q-axis components of rotor current are obtained by,

$$\begin{cases} i_{\mathrm{r-d}}^* = \left(k_{\mathrm{pP}} + \frac{k_{\mathrm{iP}}}{s}\right)\left(P_{\mathrm{s}}^* - P_{\mathrm{s}}\right), \\ i_{\mathrm{r-q}}^* = -\left(k_{\mathrm{pQ}} + \frac{k_{\mathrm{iQ}}}{s}\right)\left(Q_{\mathrm{s}}^* - Q_{\mathrm{s}}\right), \end{cases} \tag{12}$$

where subscript "P" represents the variables of stator active power control loop.

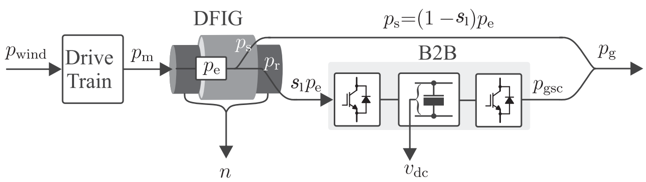


Fig. 5. Power propagation route.

The inner loop of RSC controls the rotor current and its output is the rotor reference voltage. Corresponding to the relationship between the rotor current and voltage, a PI controller can also be used to control the rotor current, which can be expressed as,

$$\boldsymbol{U}^*_{\rm r-dq} = \left(k_{\rm pi1} + \frac{k_{\rm ii1}}{s}\right)\left(\boldsymbol{I}^*_{\rm r-dq} - \boldsymbol{I}_{\rm r-dq}\right) + j\omega_{\rm slip}\sigma L_{\rm r}\boldsymbol{I}_{\rm r-dq} + \frac{L_{\rm m}}{L_{\rm s}}\frac{\omega_{\rm slip}}{\omega_0}\boldsymbol{U}_{\rm s-dq}, \quad (13)$$

where $k_{\rm pi1}$ and $k_{\rm ii1}$ represent the proportional gain and integral gain of current controller, respectively, $\boldsymbol{I}_{\rm r-dq}$, $\boldsymbol{U}_{\rm r-dq}$ and $\boldsymbol{U}_{\rm s-dq}$ are the vectors of rotor current, rotor voltage and stator voltage in the dq reference frame, $\omega_{\rm slip}$ is the slip angular frequency, ω_0 is grid voltage angular frequency, $L_{\rm m}$, $L_{\rm r}$ and $L_{\rm s}$ are the mutual, rotor and stator inductance, respectively, and $\sigma = 1 - L_{\rm m}^2/(L_{\rm s}L_{\rm r})$, $\boldsymbol{I}_{\rm r-dq} = i_{\rm r-d} + ji_{\rm r-q}$, $\boldsymbol{U}_{\rm r-dq} = u_{\rm r-d} + ju_{\rm r-q}$, $\boldsymbol{U}_{\rm s-dq} = u_{\rm s-d} + ju_{\rm s-q}$.

The control scheme of GSC also contains an outer loop and an inner loop. The outer loop maintains the DC-link voltage constant, and therefore undertakes the rotor power transmission of the DFIG. Based on the grid voltage orientation, the reference of grid current can be expressed as,

$$i^*_{\rm g-d} = \left(k_{\rm pdc} + \frac{k_{\rm idc}}{s}\right)\left(v^*_{\rm dc} - v_{\rm dc}\right), \quad (14)$$

where $v_{\rm dc}$ is the DC-link voltage, subscript "dc" represents the variables of DC-link voltage loop. The inner loop of GSC controls the grid current,

$$\boldsymbol{U}^*_{\rm g-dq} = \left(k_{\rm pi2} + \frac{k_{\rm ii2}}{s}\right)\left(\boldsymbol{I}^*_{\rm g-dq} - \boldsymbol{I}_{\rm g-dq}\right) + j\omega_0 L_{\rm f}\boldsymbol{I}_{\rm g-dq} + \boldsymbol{U}_{\rm g-dq}, \quad (15)$$

where $\boldsymbol{U}_{\rm g-dq}$ and $\boldsymbol{I}_{\rm g-dq}$ are the grid voltage and current vectors of GSC in the dq reference frame, $k_{\rm pi2}$ and $k_{\rm ii2}$ represent the proportional gain and integral gain of current controller, respectively, $L_{\rm f}$ is the filter inductance, $\boldsymbol{I}_{\rm g-dq} = i_{\rm g-d} + ji_{\rm g-q}$, $\boldsymbol{U}_{\rm g-dq} = u_{\rm g-d} + ju_{\rm g-q}$.

III. Impacts of Multi-Physical System on Power Propagation Characteristics

The power propagation paths from wind energy to electricity grids are shown in Fig. 5. The WT converts the wind power $P_{\rm w}$ into mechanical power $P_{\rm m}$ through the drive train system, and then the generator converts $P_{\rm m}$ into electromagnetic power $P_{\rm e}$, which includes the stator power $P_{\rm s}$ and rotor power $P_{\rm r}$. Although $P_{\rm s}$ is fed into the electricity grid directly, it is controlled by RSC and hence propagation characteristics of $P_{\rm s}$ is affected by the control of RSC. On the other hand, $P_{\rm r}$ is fed into the electricity grid through a back-to-back converter and hence the propagation of $P_{\rm r}$ is affected by the back-to-back converter system and the corresponding control schemes as well.

A. Power Propagation Characteristics in 2-Mass Drive Train System of WT

Based on (1)-(7), (10) and Fig. 3, the transfer function of the two-mass drive train system can be expressed as,

$$\begin{cases} G_{\rm dt}(s) = \frac{\Delta\omega_1}{\Delta T_{\rm w} - \Delta T_{\rm m}} = \frac{1}{J_1 s + D_1} \\ G_{\rm rt}(s) = \frac{\Delta\omega_{\rm r}}{\Delta T_{\rm m} - \Delta T_{\rm e}} = \frac{1}{J_2 s + D_2} \end{cases} \quad (16)$$

where $T_{\rm e}$ is the electromagnetic torque, $T_{\rm m}$ is the torsional torque related to the equivalent shaft, ω_1 and $\omega_{\rm r}$ are the angular speed of the equivalent mass-1 and mass-2, respectively, and they are related as,

$$\Delta T_{\rm m} = \left(\frac{K_{\rm eq}}{s} + D_{\rm eq}\right)\left(\Delta\omega_1 - \Delta\omega_{\rm r}\right), \quad (17)$$

Based on (17), $\Delta\omega_1$ can be represented by $\Delta\omega_{\rm r}$ as

$$\Delta\omega_1 = \Delta\omega_{\rm r} + \frac{\Delta T_{\rm m}}{\frac{K_{\rm eq}}{s} + D_{\rm eq}}. \quad (18)$$

By substituting (18) into (16) and multiplying the obtained equations by $1/\omega_{\rm r}$, one can obtain the transfer functions of rotor speed fluctuations with respect to wind, mechanical and electromagnetic power fluctuations, which are shown as,

$$\begin{cases} G_{\rm DT}(s) = \frac{\Delta\omega_{\rm r}}{\Delta P_{\rm w} - G_{\rm p}\Delta P_{\rm m}} = \frac{1}{(J_1 s + D_1)\omega_{\rm r}}, \\ G_{\rm RT}(s) = \frac{\Delta\omega_{\rm r}}{\Delta P_{\rm m} - \Delta P_{\rm e}} = \frac{1}{(J_2 s + D_2)\omega_{\rm r}}, \end{cases} \quad (19)$$

where

$$G_{\rm p}(s) = 1 + \frac{J_1 s^2 + D_1 s}{D_{\rm eq} s + K_{\rm eq}}. \quad (20)$$

B. Power Propagation Characteristics of WT System Based on Rotor Speed Control

The control schemes of DFIG also have a significant impact on the propagation characteristics of power from wind energy to electricity grids.

As studied in Section II and (19), when the RSC adopts a speed loop control scheme, the control diagram composed of drive train, RSC and generator is shown in Fig. 6, where $s_{\rm l}$ is the slip of DFIG, $G_{\rm cc1}$ represents the control system and the physical circuit. The control system includes the PI controller of outer speed loop denoted as $G_{\rm os}$ and the PI controller of inner current loop denoted as $G_{\rm ic}$. The physical circuit includes the physical circuit of DFIG, $G_{\rm pc}$, and power gains, $G_{\rm k}$ and $(1 - s_{\rm l})^{-1}$, which denotes the transformation from the rotor current

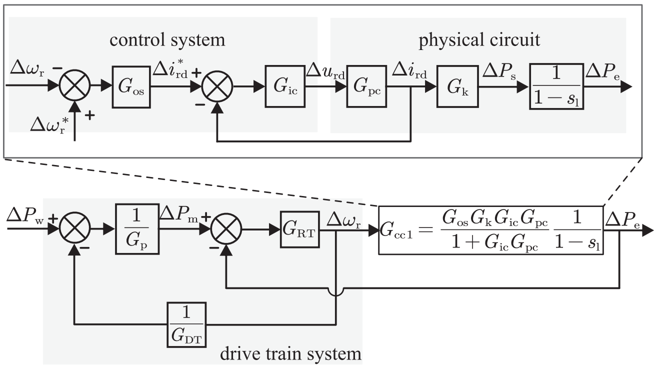


Fig. 6. Flow chart of the drive train, generator and RSC adopting speed control scheme.

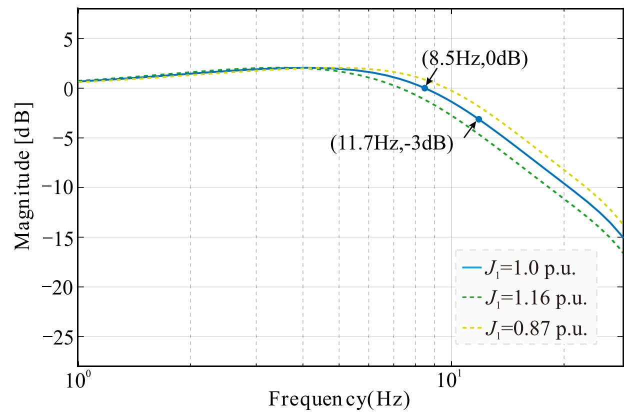


Fig. 7. Bode plot of H_{we}.

TABLE II
CONTROL PARAMETERS OF THE BACK-TO-BACK CONVERTER

Description	Symbols	Values
Speed controller (RSC)	$k_{\mathrm{p}\omega}, k_{\mathrm{i}\omega}$	7500, 20325
Reactive power controller (RSC)	$k_{\mathrm{pQ}}, k_{\mathrm{iQ}}$	0.0071, 0.0237
Active power controller (RSC)	$k_{\mathrm{pP}}, k_{\mathrm{iP}}$	0.0071, 0.0237
Current controller (RSC)	$k_{\mathrm{pi1}}, k_{\mathrm{ii1}}$	0.0179, 0.1917
DC voltage controller (GSC)	$k_{\mathrm{pdc}}, k_{\mathrm{idc}}$	0.0225, 0.0451
Current controller (GSC)	$k_{\mathrm{pi2}}, k_{\mathrm{ii2}}$	0.1110, 5.5549

to the electromagnetic power. They can be expressed as,

$$\begin{cases} G_{\mathrm{os}} = -\left(k_{\mathrm{p}\omega} + \frac{k_{\mathrm{i}\omega}}{s}\right), \ G_{\mathrm{ic}} = k_{\mathrm{pi1}} + \frac{k_{\mathrm{ii1}}}{s} \\ G_{\mathrm{pc}} = \frac{1}{s\sigma L_{\mathrm{r}} + R_{\mathrm{r}} + (L_{\mathrm{m}}/L_{\mathrm{s}})^2 R_{\mathrm{s}}} \frac{1}{1.5T_{\mathrm{s}}s + 1} \\ G_{\mathrm{k}} = \frac{3}{2}\frac{L_{\mathrm{m}}}{L_{\mathrm{s}}} U_{\mathrm{sd}} \\ G_{\mathrm{cc1}} = \frac{G_{\mathrm{os}} G_{\mathrm{k}} G_{\mathrm{ic}} G_{\mathrm{pc}}}{1 + G_{\mathrm{ic}} G_{\mathrm{pc}}} \frac{1}{1 - s_{\mathrm{l}}} \end{cases} \tag{21}$$

where T_{s} is the switching period of the converter, R_{r} is the rotor resistance, and R_{s} is the stator resistance.

Based on Fig. 6, the transfer function from fluctuation of wind power ΔP_{w} to electromagnetic power ΔP_{e} can be derived,

$$H_{\mathrm{we}} = \frac{G_{\mathrm{RT}} G_{\mathrm{cc1}} G_{\mathrm{DT}}}{G_{\mathrm{p}} G_{\mathrm{DT}} \left(1 + G_{\mathrm{RT}} G_{\mathrm{cc1}}\right) + G_{\mathrm{RT}}}. \tag{22}$$

Based on the parameters of WT system listed in Tables I and II, the Bode diagram of H_{we} is shown as the blue solid line in Fig. 7, where the bandwidth is 11.7 Hz. The fluctuations of ΔP_{w} is amplified slightly when the frequency is less than 8.5 Hz, while the frequency of wind power fluctuations are attenuated by drive train significantly with the increase of the fluctuation frequency. Dash lines in Fig. 7 also show that as the inertia of the drive train is increased from 0.87 p.u. to 1.16 p.u., the crossover frequency decreases from 9.6 Hz to 7.3 Hz, and bandwidth decrease from 13.1 Hz to 10.2 Hz.

C. Power Propagation Characteristics of WT System Based on Stator Active Power Control

The control block diagram of the RSC when using the stator active power loop control is shown in Fig. 8, where ΔP_{s}^* is the reference of ΔP_{s}, G_{op} represents the PI controller of outer power loop, G_{cc2} is the open-loop transfer function of the active power, and they can be expressed as,

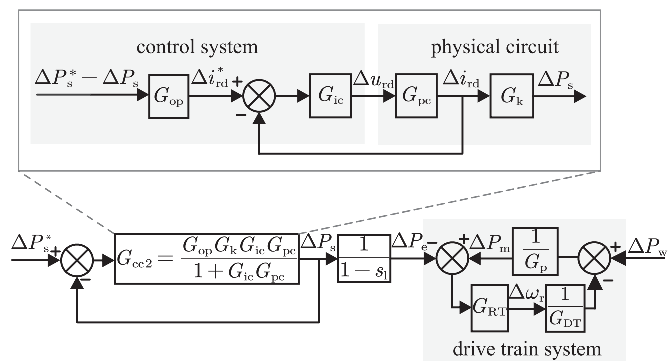


Fig. 8. Flow chart of the drive train, generator and RSC adopting power control scheme.

$$\begin{cases} G_{\mathrm{op}} = k_{\mathrm{pP}} + \frac{k_{\mathrm{iP}}}{s}, \\ G_{\mathrm{cc2}} = \frac{G_{\mathrm{op}} G_{\mathrm{k}} G_{\mathrm{ic}} G_{\mathrm{pc}}}{1 + G_{\mathrm{ic}} G_{\mathrm{pc}}}. \end{cases} \tag{23}$$

As illustrated in the drive train module of Fig. 8, the input wind power fluctuations ΔP_{w} mainly cause oscillations in the mechanical power and rotor speed of the drive train. This indicates that, under stator active power control, the mechanical drive train system effectively absorbs the fluctuating wind power through rotor speed variations, buffering it before it impacts the electrical system, thus significantly suppressing its propagation to the grid. This is further validated by the comparison of rotor speed under speed control and power control schemes, as shown in Fig. 9, where the rotor speed fluctuation under the power control scheme is noticeably greater than that under the speed control scheme.

As shown in Fig. 8, due to the closed-loop control of stator active power P_{s}, fluctuations in the input wind power do not directly propagate to P_{s}, resulting in minimal fluctuations in it. However, due to the coupling effect between rotor speed fluctuations and slip s_{l}, there are minimal disturbances in electromagnetic power. When the rotor speed fluctuates within a certain range, the slip s_{l} fluctuates synchronously, leading to fluctuations and 3p harmonics in the electromagnetic power. And due

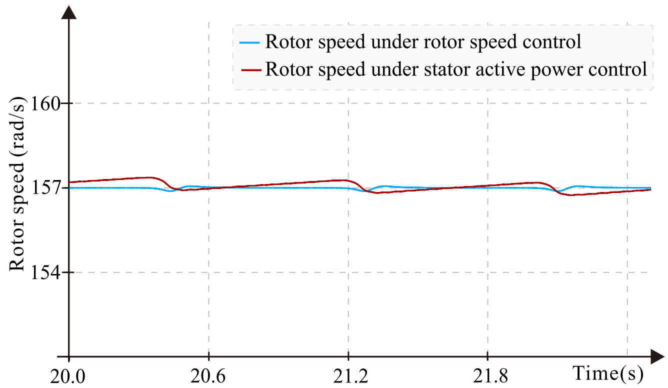


Fig. 9. Rotor speed under rotor speed control and stator active power control.

to the large inertia of the mechanical drive train system, the rotor speed fluctuations caused by absorbed fluctuating wind power are relatively small. As shown in Fig. 9, the rotor speed fluctuates between 156.7 rd/s and 157.3 rd/s, with slip fluctuations within 0.23%, both of which are small. Consequently, the fluctuations in electromagnetic power are minor. Therefore, while fluctuations in electromagnetic power caused by fluctuating wind power exist, they are minimal, indicating that the fluctuating wind power has an insignificant impact on the electrical system and is almost not propagated to the grid side under stator active power control.

IV. Electricity Grid Performances Analysis

By neglecting the power losses in DFIG and considering the power balance, it can be obtained as,

$$\begin{cases} P_{\mathrm{m}} = P_{\mathrm{e}} = P_{\mathrm{s}} + P_{\mathrm{r}},\ \ P_{\mathrm{s}} = (1-s_1)\,P_{\mathrm{e}}, \\ P_{\mathrm{g}} = P_{\mathrm{gsc}} + P_{\mathrm{s}}, \qquad P_{\mathrm{r}} = s_1 P_{\mathrm{e}}. \end{cases} \tag{24}$$

where P_{g} is the grid active power and P_{gsc} is the ac side active power of GSC.

Based on (6) and (7), when the mechanical rotational speed of WT is smooth and constant, the fluctuating mechanical torque $\tilde{T}_{\mathrm{m}}$ and power $\tilde{P}_{\mathrm{m}}$ have the same 3p frequency, caused by wind shear and tower shadow effects. Furthermore, based on (24), it yields that P_{e}, P_{r} and P_{s} have the 3p components ($\tilde{P}_{\mathrm{e}}$, $\tilde{P}_{\mathrm{r}}$ and $\tilde{P}_{\mathrm{s}}$) as well.

In the back-to-back converter, by neglecting power losses, the formula of dc-link input power, which is equal to P_{r}, output power P_{gsc} and voltage v_{dc} can be expressed as,

$$P_{\mathrm{r}} - P_{\mathrm{gsc}} = v_{\mathrm{dc}} C_{\mathrm{dc}} \frac{dv_{\mathrm{dc}}}{dt}, \tag{25}$$

where C_{dc} is the dc capacitance.

When the dc-link voltage is controlled to be constant, based on (25), it yields that P_{gsc} has the fluctuation with the same 3p frequency as well. And according to (24), P_{g} has the same 3p frequency fluctuation. Similar to (6), the grid power can be expressed as,

$$P_{\mathrm{g}} = \bar{P}_{\mathrm{g}} + \tilde{P}_{\mathrm{g}}. \tag{26}$$

and P_{g} can be calculated by,

$$P_{\mathrm{g}} = 3U_{\mathrm{g}} I_{\mathrm{g}} \cos\varphi, \tag{27}$$

where U_{g} and I_{g} are the root mean square (RMS) values of the three-phase voltage and current on the grid side respectively, φ represents the phase difference between current and voltage.

Based on (26) and (27), when grid voltage remains relatively constant, the effective value of the three-phase grid current can be expressed as,

$$I_{\mathrm{g}} = \frac{1}{3}\frac{1}{U_{\mathrm{g}}\cos\varphi}\left(\bar{P}_{\mathrm{g}} + \tilde{P}_{\mathrm{g}}\right) = \bar{I}_{\mathrm{g}} + \tilde{i}_{\mathrm{g}} \tag{28}$$

where

$$\bar{I}_{\mathrm{g}} = \frac{1}{3}\frac{1}{U_{\mathrm{g}}\cos\varphi}\bar{P}_{\mathrm{g}},\ \tilde{i}_{\mathrm{g}} = \frac{1}{3}\frac{1}{U_{\mathrm{g}}\cos\varphi}\tilde{P}_{\mathrm{s}}. \tag{29}$$

The fluctuation frequency is assumed to be $\tilde{\omega}$, and it can be obtained that,

$$\tilde{i}_{\mathrm{g}} = \tilde{I}_{\mathrm{g}} \cos\tilde{\omega}t. \tag{30}$$

Taken the phase-A current as an example, when the initial phase is assumed to be zero, based on (28) and (30), it can be obtained that,

$$i_{\mathrm{ga}} = \sqrt{2}\bar{I}_{\mathrm{g}}\cos(\omega_0 t - \varphi) + \sqrt{2}\tilde{I}_{\mathrm{g}}\cos(\omega_0 t - \varphi)\cos\tilde{\omega}t. \tag{31}$$

The second term in (31) is the disturbance of phase-A current, and can be computed by,

$$\begin{aligned} &\sqrt{2}\tilde{I}_{\mathrm{g}}\cos(\omega_0 t - \varphi)\cos\tilde{\omega}t \\ &\quad = \frac{\sqrt{2}\tilde{I}_{\mathrm{g}}}{2}\left[\cos\left(\omega_0 + \tilde{\omega}\right)t + \cos\left(\omega_0 - \tilde{\omega}\right)t\right]. \end{aligned} \tag{32}$$

It shows that in (32) the grid current contains sub-synchronous $(\omega_0 - \tilde{\omega})$ and complementary super-synchronous $(\omega_0 + \tilde{\omega})$ components.

V. Simulation Validation of Power Propagation Characteristics

The impacts of WT drive train and control schemes of DFIG on the power propagation characteristics are validated by the simulation results of DFIG-based WT system carried out in PLECS, with the mechanical parameters and control parameter listed in Table I [29] and Table II.

A. Characteristics of Wind Power

The waveforms of wind power in the time domain and frequency domain are shown in Fig. 10. It can be observed that the wind power exhibits three occurrences of minimum values within a rotation period due to the effects of wind shear and tower shadow. It results in the wind power experiencing the fluctuation with $3n$p frequencies as shown in Fig. 10(b), which is consistent with the theoretical analysis in (6). The frequency of 3p fluctuation is 1.2 Hz, which is the harmonics propagated from the blade passing frequency.

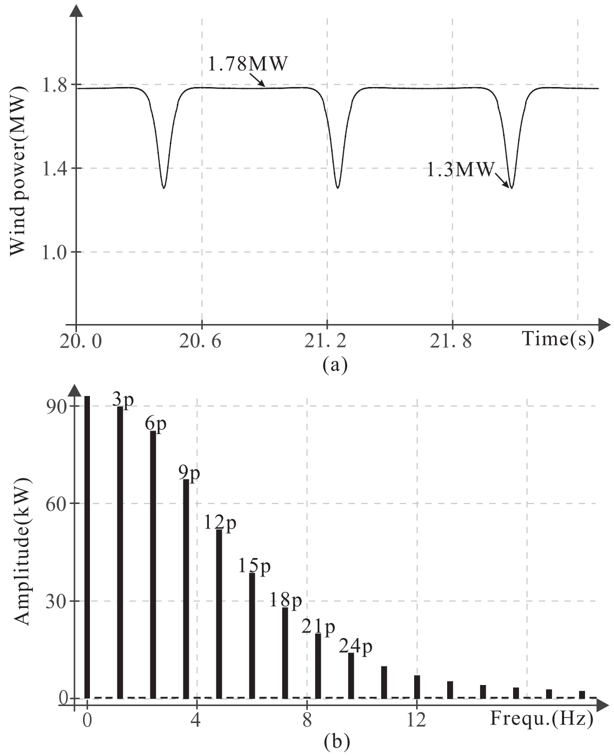


Fig. 10. Wind power characteristics, (a) wind power in time-domain and (b) its spectra.

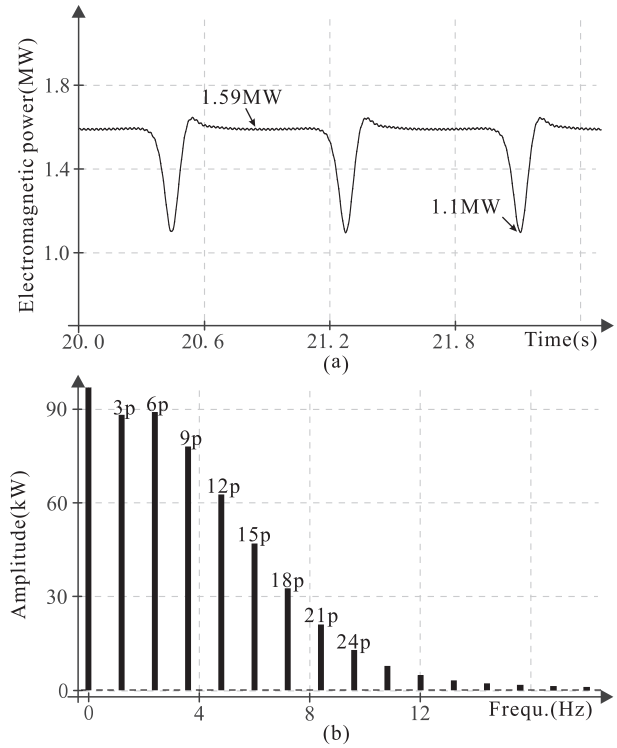


Fig. 11. The power propagation characteristics of WT system adopting speed control scheme, (a) the electromagnetic power in time-domain and (b) its frequency spectra.

B. Power Propagation Characteristics of WT System Based on Rotor Speed Control

The power propagation characteristics of the WT system with speed control scheme are shown in Fig. 11. In comparison of Figs. 10 and 11, it can be seen that P_{w} and P_{e} have the same fluctuating frequency, the fluctuation frequencies above 10 Hz in wind power P_{w} extracted by WT are attenuated as compared to the electromagnetic power P_{e}. In contrast, fluctuations below 10 Hz are propagated to the electrical system with minimal attenuation or even slight amplification, matching with the theoretical analysis in Fig. 7.

Based on the simulation results in Figs. 10 and 11, the frequency response of $\Delta P_{\mathrm{e}}/\Delta P_{\mathrm{w}}$ can be fitted, as shown in Fig. 12. It should be noted that the simulation is conducted with the six-mass drive train model as shown in Fig. 2, while the simplified two-mass model as shown in Fig. 3 is adopted in theoretical analysis. It can be observed from Fig. 12 that the fitting bode diagram based on simulation results closely matches the theoretical bode plot, validating the accuracy of the theoretical analysis in Section III and the effectiveness of the simplified drive train model.

C. Power Propagation Characteristics of WT System Based on Stator Active Power Control

For comparison, the power propagation characteristics of the WT system with active power control scheme are simulated in Fig. 13, where the harmonic components of the electromagnetic power at frequencies of $3n$p are minimized. The value is approximately 2% of the corresponding harmonic components of wind power. The $3n$p harmonics are generated by the slip s_1, which is affected by the rotor speed without direct control of the rotor speed in this scheme. It is consistent with the analytical result in Fig. 8.

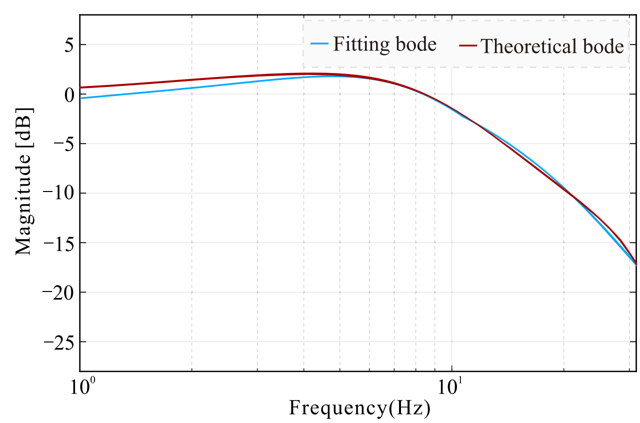


Fig. 12. Propagation characteristics of WT system.

The propagation of fluctuating wind power is significantly suppressed compared to the speed control scheme, resulting in a notable reduction in its impact on the electromagnetic power and electricity grids. It demonstrates the significant influence of the generator control scheme on the propagation characteristics of fluctuating power.

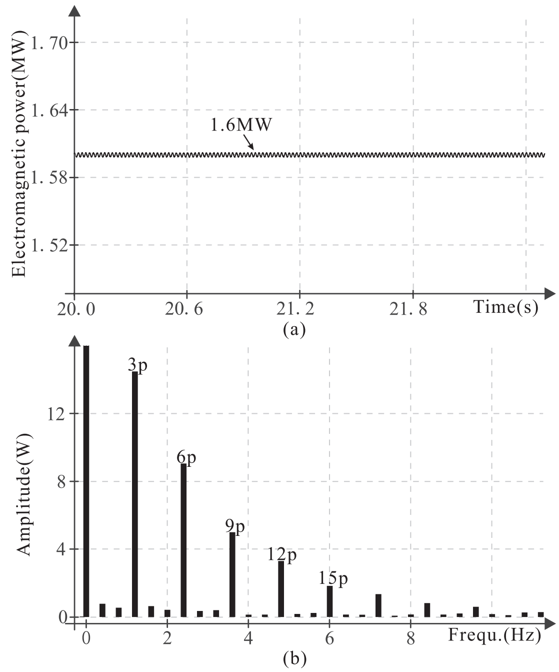


Fig. 13. The power propagation characteristics of WT system adopting power control scheme, (a) the electromagnetic power in time-domain and (b) its frequency spectra.

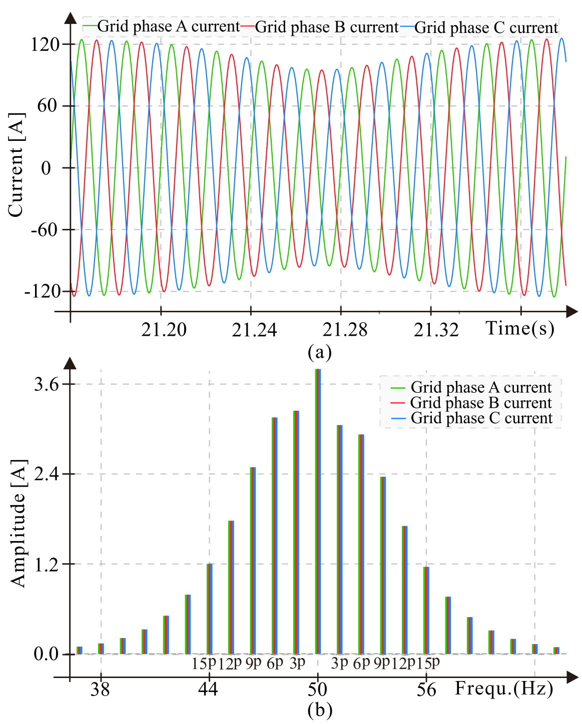


Fig. 14. Results of (a) three-phase currents of grid and (b) its frequency spectra.

D. Impacts on Electricity Grid Performances

The performances of three-phase grid currents when adopting speed control scheme are shown in Fig. 14, where the grid currents contain the harmonics with frequencies of $50 \pm 3np$ Hz

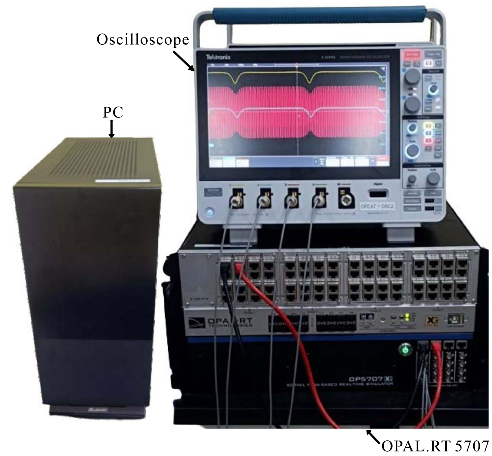


Fig. 15. Photograph of OPAL-RT 5707 platform.

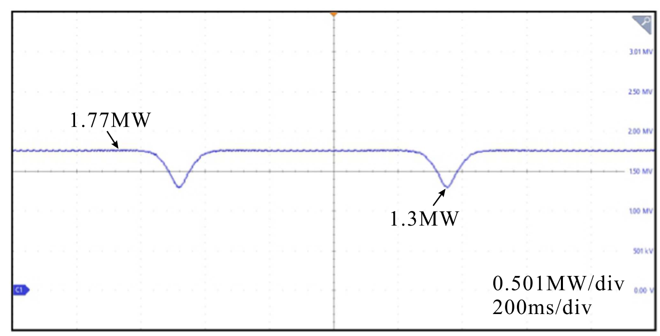


Fig. 16. Wind power extracted by WT considering the effects of wind shear and tower shadow, based on the OPALRT-LAB 5707 platform.

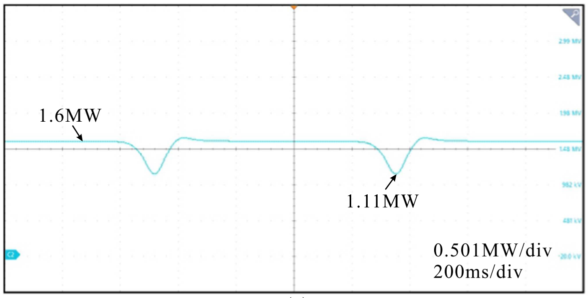
caused by wind shear and tower shadow. They are all consistent with the theoretical results in Section IV.

VI. Real-Time Digital Simulator-Based Verification

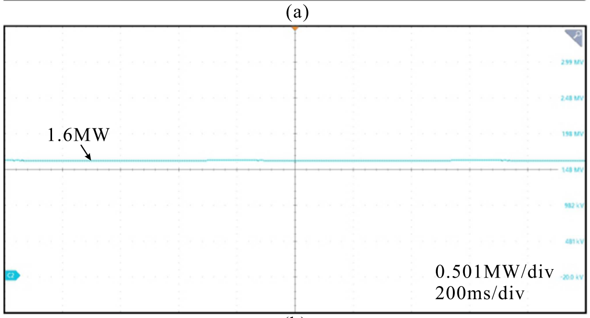
Based on the parameters listed in Tables I and II, the OPALRT-LAB 5707 platform-based results are as shown in Fig. 15, to verify the theoretical analysis and simulation results.

The wind speed at the hub height is set to 11.2 m/s, in accordance with the simulation. Fig. 16 depicts the waveform of wind power extracted by WT considering the effects of wind shear and tower shadow. It can be seen that the steady-state value of wind power is 1.77 MW and power fluctuations occur with a minimum value of 1.3 MW, which is similar to that in Fig. 10(a).

As discussed in Section III, control strategies significantly influence the power propagation characteristics of fluctuating wind power. The performances of electromagnetic powers and grid currents when adopting speed control scheme and power control scheme, are compared in Figs. 17 and 18. When the speed control scheme is used, it can be seen from Fig. 17(a) that the steady-state value of electromagnetic power is 1.6 MW and power fluctuations occur with a minimum value of 1.11 MW when using speed control scheme, and the grid currents as shown in Fig. 18(a) exhibit fluctuations, leading to the presence of harmonics at frequencies of $50 \pm 3np$ Hz in the electricity

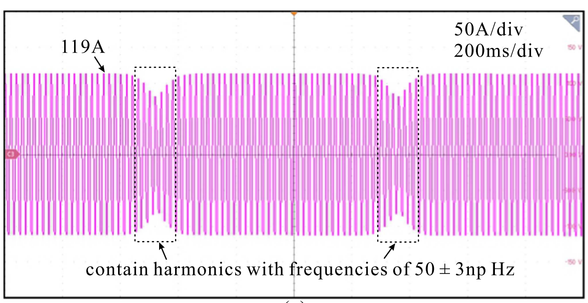


(a)

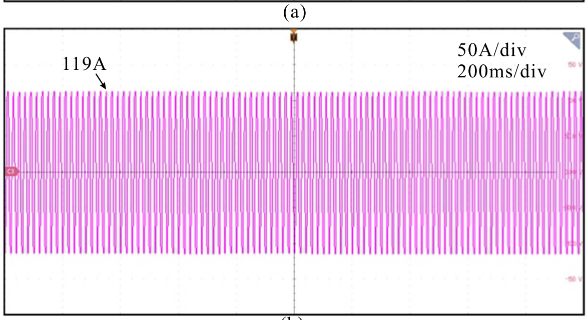


(b)

Fig. 17. Results of electromagnetic power with (a) speed control scheme and (b) power control scheme, based on the OPALRT-LAB 5707 platform.



(a)



(b)

Fig. 18. Results of grid current with (a) speed control scheme and (b) power control scheme, based on the OPALRT-LAB 5707 platform.

grids. In contrast, when using power control scheme, the electromagnetic power and the amplitude of grid current almost remain constant at 1.6 MW and 119 A, exhibiting minimal fluctuations regardless of the variations in wind power, as shown in Figs. 17(b) and 18(b). The results are all satisfied with the theoretical analysis in Sections III and IV.

VII. Conclusion

In this paper, the multi-physical domain model of wind turbine system composed of the aerodynamical system, mechanical drive train, doubly-fed induction generator, and back-to-back converter as well as its control schemes (power control and speed control of the generator) is derived to study its impacts on the power propagation and frequency coupling characteristics from wind energy to electricity grids. The wind shear and tower shadow effects result in the wind power experiencing 3p fluctuations, which are transformed into $50 \pm 3np$ Hz components in the ac grids. The drive train of the wind turbine exhibits the low-pass characteristics with 11.7 Hz bandwidth, and the bandwidth is decreased from 13.1 Hz to 10.2 Hz with the studied 2 MW wind turbine as the drive train inertia is increased from 0.87pu to 1.16pu. The control schemes of doubly-fed induction generator have a significant impact on the propagation and frequency coupling characteristics. The fluctuations of wind power are attenuated at the frequencies higher than the crossover frequency around 8.5 Hz by the wind turbine system, while are propagated into electricity grids with minimal attenuation or even slight amplification in the frequency band less than 8.5 Hz when adopting the speed control scheme. However, when using power control scheme, the fluctuations of wind power across the entire frequency range are almost not propagated to the grid and buffered by the rotor due to the closed-loop control of the electromagnetic power. The test results carried out in the PLECS and the OPALRT-LAB 5707 platform are consistent with the theoretical analyses.

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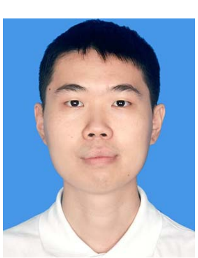
Lulan Yin received the B.S. degree in hydraulic and hydropower engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2022. She is currently working toward the M.S. degree in electrical engineering with Harbin Institute of Technology, Shenzhen, China. Her research focuses on the oscillation mechanism of wind power generation systems.



Weihao Hu received the B.S. degree in electrical engineering and automation from the Harbin Institute of Technology, Shenzhen, China, in 2023. He is currently working toward the M.S. degree in electrical engineering with Harbin Institute of Technology, Shenzhen, China. His research focuses on the stability issues of renewable energy generation.

Rongwu Zhu (Senior Member, IEEE) received the B.Eng. degree in electrical engineering from Nanjing Normal University, Nanjing, China, in 2007, and the Ph.D. degree in energy technology from the Department of Energy Technology, Aalborg University, Aalborg, Denmark, in 2015. From 2011 to 2012, he was a Guest Researcher with Aalborg University. From 2016 to 2021, he was a group leader of renewable power grid and Senior Researcher with the Chair of Power Electronics, Christian-Albrechts University of Kiel, Kiel, Germany. He is currently a Full Professor, Director of Department of Electrical Engineering and Vice Dean of School of Robotics and Advanced Manufacturing with Harbin Institute of Technology, Shenzhen, China. He is also a Director of the HITsz-CYGsr joint lab of modern distribution power system and intelligent equipment. He has authored or coauthored more than 140 technical papers (more than 1/3 of them in international peer-reviewed journals/magazine), 1 book chapter, and 25 granted or pending patents. He won the third prize of the "Data Element-x" China national competition 2024. His research interests include advanced control and grid integration of renewable energy resources, autonomous control of renewables-dominated grids, AI technology for power electronics and electricity grids, and digitization and talkative power conversion. He served as an Editor of the *International Transactions on Electrical Energy System*, *Associate Editor of the IET Renewable Power Generation*, Associate Editor of IEEE Open Journal of Power Electronics, and Guest Associate Editor of the IEEE Journal of Emerging and Selected Topics in Power Electronics, Guest Editor-in-Chief of CSEE Journal of Power and Energy Systems, and Committee Chairperson for several International Conferences.

Marco Liserre (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in electrical engineering from the Politecnico di Bari, Bari, Italy, in 1998 and 2002 respectively. He has been an Associate Professor with the Politecnico di Bari, since 2012, and has been a Professor of reliable power electronics with Aalborg University. Alborg, Denmark. Since 2013, he has been a Full Professor and the Chair of power electronics with the University of Kiel, Kiel,Germany. He has been offered and declined professorships with several universities. He has authored or coauthored more than 700 technical papers (1/3 of them in international refereed journals), one book and seven granted patents (four with companies). His works have received more than 50,000 citations. He was selected as an Highly Cited Researcher in the field of Engineering (Clarivate Web of Science) from 2014 to 2021. Several of his students (M.Sc., Ph.D. and Post-docs) are in leading positions in industry and universities worldwide. In 2023, he joined the Fraunhofer ISIT on a part-time basis as Deputy Director and Director of the new division "Electronic Energy Systems", and of the Kiel branch of the Fraunhofer ISIT. From 2025 he is acting Director of Fraunhofer ISIT. He is a member of IAS, PELS, PES and IES. He has served all these societies in various capacities. In PELS, he is Co-Editor of the IEEE Open Access Journal in Power Electronics and Technical Committee Chairman of the Committee on Electronic Power Grid Systems. He has co-chaired several IEEE conferences being several times Chairman. He was the recipient of 16 awards from IEEE, PCIM and EPE-PEMC, including the prestigious 2018 IEEE-IES Mittelmann Achievement Award and 2023 IEEE-PELS R. David Middlebrook Achievement Award. In 2023, he was awarded the title of "Ufficiale" by the President of the Italian Republic. In 2025, he will be Chairman of Powertech 2025 in Kiel.

Zhe Chen (Fellow, IEEE) received the B.Eng. and M.Sc. degrees in electrical engineering from the Northeast China Institute of Electric Power Engineering, Jilin City, China, and the Ph.D. degree in Power and Control, from the University of Durham, England. Since 2002 he has been a full Professor with the Department of Energy Technology, Aalborg University, Denmark. He is the leader of Wind Power System Research team with the Department of Energy Technology, Aalborg University. His main research interests are renewable energy, power electronics, power systems, multi energy systems and AI applications in energy systems. He has led many research projects and has more than 1000 publications in these technical fields. Dr. Chen is an Associate Editor of IEEE Transactions on Power Electronics, a member of editorial boards for many international journals, a Fellow of the Institution of Engineering and Technology (IET, London, U.K.), and a Chartered Engineer in the U.K.